

Sandra Maritza Principe

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Skills

Programming Languages: C++, Assembly Language(MIPS), Java

Tools: UNIX Commands, Microsoft Excel (data formatting, statistical functions), SQL

Technical Concepts: Digital Circuits, Data Structures (trees, stacks, arrays, queues), Binary & Hexadecimal Systems

Soft Skills: Communication, Time Management, Adaptability, Collaboration, Problem Solving, Attention to detail

Projects

Craps Game: C++

- Simulated craps game using random number generator for two variables
- Implements rules of craps game to determine win/loss

Median and Reverse: MIPS

- MIPS program that allows array number inputs from user and displays the median of the array and displays the array in reverse
- Uses loops, function calling, and sorting, register operations

Fahrenheit to Celsius Converter: MIPS

- MIPS program that will convert Celsius into Fahrenheit
- Incorporates user's input and uses the conversion equation $F = (C * 9/5) + 32$
- Uses integer to float conversions

3rd Place Hackathon

- Collaboration with a team of four to create solutions for four different coding challenges
- Worked on C++ programs with a partner to write a program to find numbers that disprove the given equation
- Prompt: **Suppose every odd natural number greater than 1 can be written as the sum of a prime number and 2 times a square as seen: $k = p + 2 * (n)^2$ Where k is an odd number, p is a prime, and n is some integer.**
- Main point of algorithm is using Sieve of Eratosthenes

Education

August 2023 - Present

California State University San Marcos, San Marcos – *Computer Information System*

Expected Gradation: May 2027

Coursework: Programming, Data Structures, Assembly Language